

without which you will be stuck copying code from Stack Overflow forever. To gain a thorough and formal understanding of these topics, I recommend the books ‘Linear Algebra Done Right’ by Sheldon Axler and ‘A First Course in Probability’ by Sheldon Ross. Please keep in mind that these are my personal recommendations, and there are many other textbooks available that cover these subjects in a way that may be more suitable for you.

Apart from the mathematical aspects, we have also done a lot of coding as part of the series. We did not approach the series in a programming language-agnostic way; rather, we relied mostly on Python for coding. While programming with Python, we used tools like JupyterLab and Anaconda Navigator. We covered almost all the major Python libraries and packages, such as TensorFlow, Keras, PyTorch, scikit-learn, Pandas, OpenCV, Matplotlib, etc, which are used for developing AI and machine learning-based applications. Identifying and utilising relevant Python libraries is crucial due to the vast number of options available, and I hope you will give it serious thought. In every article in this series, we also discussed the theoretical aspects of AI and machine learning. I hope that you will continue to update yourselves with the most recent developments in the AI industry.

Let us now examine the various AI and machine learning paradigms that we explored throughout this series. We provided comprehensive coverage of supervised learning and delved into the specifics of classification and regression using different Python libraries. However, to truly enhance your understanding of supervised learning, it is essential to work through more examples. While support vector machines (SVMs) are a critical component of supervised

learning, our coverage of them was minimal, and we did not engage in any theoretical discussion regarding them. As a result, I highly recommend dedicating time to master both the theory and application of SVMs.

Our coverage of unsupervised learning was relatively limited compared to supervised learning. We presented an example of how clustering could be implemented, but there is still much more to learn about unsupervised learning, even with clustering. Additionally, we entirely omitted the topic of dimensionality reduction, a crucial aspect of unsupervised learning. By doing so, we disregarded techniques such as principal component analysis (PCA) and independent component analysis (ICA). Therefore, I suggest taking the time to study these techniques in-depth.

We briefly touched on the two lesser-known paradigms of machine learning, semi-supervised learning and reinforcement learning. Acquiring knowledge in these areas could prove beneficial in the long run for your journey through AI and machine learning. Neural networks are the fundamental building blocks for AI-based applications. Learning the specifics of neural networks formally, rather than informally as we treated them, could have a tremendous impact on your understanding of AI.

As we come to the end of this 12-part series on AI, I am reminded of a quote by the famous poet W.H. Auden, “A poem is never finished, only abandoned.” Looking back, I feel the same way about this series. There were many things I could have done differently, such as adding or

excluding topics or improving the presentation. However, I am happy with the journey we have taken, which I feel has served its purpose of introducing eager learners to the field of AI. Although I may have deviated from the original plan on occasions, I have remained faithful to the goal of introducing the maths and programming behind AI-based applications, albeit less formally than originally intended. Indeed, what sets this series apart, in my opinion, is our unique approach of blending the theoretical, mathematical, and programming aspects of AI.

During our one-year journey through AI, we witnessed a growing interest in the subject from the general public, which reached its peak with the introduction of ChatGPT. As an example of this, ChatGPT was featured on the cover page of the March 2023 issue of *Frontline* magazine. I cannot recall any other software or tool making such headlines in such a short amount of time. It would not be surprising if ChatGPT is chosen as *Time* magazine’s ‘Person of the Year’ for 2023, much like ‘The Computer’ was chosen in 1982.

As we end this series at a time when there is a major breakthrough in the field of AI, I offer these parting words of advice: formally learn the mathematical concepts introduced in this series, delve deeper into the Python libraries, and start developing a project that interests you which is of intermediate size and complexity, in order to practice the skillsets you have gained through this series.

I end with the hope that the series has been beneficial to you, and I extend my best wishes for an enjoyable and thrilling journey in the domain of AI. **END** 🐧

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